

SUMMARY

- **Research Interests:** My research focuses on the **mechanical design, motion control, and navigation of legged robots**, with an emphasis on robotic systems operating in complex and unstructured field environments. I am particularly interested in **bio-inspired hexapod robots, robot locomotion, motion planning, robot perception**, and the integration of sensing and control for robust field deployment.
- **Highlight:** Research experience in legged robot design, control, perception, and navigation; Hands-on experience in designing and building robotic hardware systems from the ground up; Proficient in MATLAB, Python, ROS2, SolidWorks, and robot simulation; Strong interdisciplinary background in mechanical design, control systems, sensing, and agricultural robotics.
- **Relevant Courses:** Computer Vision (A+), Sensors and Controls (A+), Fundamentals of Engineering Mathematics (A+), Experimental Statistics for Engineers (A), Advanced Robotics (A), University Physics II-1 (A), Theoretical Mechanics (A-), Environmental Analytics Modeling (A-), Structural Dynamics of Aircraft (A-),

EDUCATION

- **Cornell University** Ithaca, NY, United States
Ph.D. student in Biological and Environmental Engineering Since Jan. 2026
- **North Carolina State University** Raleigh, NC, United States
Ph.D. student in Biological and Agricultural Engineering Aug. 2023 - Jan. 2026
GPA: 3.68/4.0
- **Xi'an Jiaotong University** Xi'an, Shaanxi, China
Bachelor's of Engineering - Flight Vehicle Design and Engineering Sep. 2019 - Jun. 2023
GPA: 3.47/4.0, Rank: 5/62

PUBLISHED PAPER

- **Development of A Bionic Hexapod Robot with Adaptive Gait and Clearance for Enhanced Agricultural Field Scouting:**
Zhenghua Zhang, Weilong He, Fan Wu, Lina Maria Quesada-Ocampo, Lirong Xiang*
Frontiers in Robotics and AI, 11:1426269. <https://doi.org/10.3389/frobt.2024.1426269>
- **High-Throughput Robotic Phenotyping for Quantifying Tomato Disease Severity Enabled by Synthetic Data and Domain-Adaptive Semantic Segmentation:**
Weilong He, Xingjian Li, Zhenghua Zhang, Yuxi Chen, Jianbo Zhang, Dilip R Panthee, Inga Meadows, Lirong Xiang*
Journal of Field Robotics, 42: 657-678. <https://doi.org/10.1002/rob.22490>

PAPER IN PROGRESS

- **AgHexaNav: A Hierarchical Navigation Framework for Hexapod Robots in Vegetation-Dense Agricultural Environments:**
Zhenghua Zhang, Qihang Li, Prem Raj, Weilong He, Shengdao Du, Peng Gao, Lirong Xiang*
Manuscript in revision for resubmission to *IEEE Robotics and Automation Letters (RA-L)*
- **Navigating Cluttered Vine Crop Fields with an Adjustable Clearance Hexapod Robot and Energy-Aware Path Planning:**
Zhenghua Zhang, Pengyao Xiel, Fan Wu, Weilong He, Shengdao Du, Lirong Xiang*
Under review at *Journal of Field Robotics*
- **Towards Accurate Mushroom Pose Estimation: Integrating Synthetic Data Scaling and Multi-View Attention Fusion:**
Fan Wu, Zhenghua Zhang, Wenqiao Yuan*
In Submission to *Computers and Electronics in Agriculture*
- **MIR3D: Monocular Infrared Geometry Estimation for Autonomous Robotic Mushroom Harvesting in Dark Environments:**
Fan Wu, Zhenghua Zhang, Wenqiao Yuan*
In Submission to *IEEE Robotics and Automation Letters (RA-L)*

RESEARCH EXPERIENCE

- Cornell University**
Graduate Research Assistant, Supervised by Dr. Lirong Xiang

Ithaca, NY, USA
Since Jan. 2026

Vision-Guided Precision Weed Spraying Using a Heterogeneous Leg-Arm Hexapod Robot [Ongoing Work]
 - Led the mechanical design of a compact , heterogeneous 5-DOF leg-arm spraying module integrated with a custom-built hexapod robot platform.
 - Developed the nozzle accuracy calibration, straight-jet spray modeling, and the leg-arm control strategy for precision weed spraying.
 - Leading outdoor experiments and data collection to evaluate spray accuracy and chemical-use reduction.
- North Carolina State University**
Graduate Research Assistant, Supervised by Dr. Wenqiao Yuan

Raleigh, NC, USA
July. 2025 - Jan. 2026

Mushroom Growth Monitoring and 3D Perception
 - Contributed to the mushroom growth monitoring and 3D perception project by leading hardware design, system assembly, and hardware debugging for the imaging and robotic sensing setup.
- North Carolina State University**
Graduate Research Assistant, Supervised by Dr. Lirong Xiang

Raleigh, NC, USA
Aug. 2023 - July. 2025

Navigating Cluttered Vine Crop Fields with an Adjustable-Clearance Hexapod Robot and Crop-Aware Navigation
 - Led the redesign and upgrade of the previously developed hexapod robot into a 24-DOF adjustable-clearance platform with an expanded body-clearance range for crop-safe navigation.
 - Developed a crop-aware navigation framework using U-Net semantic segmentation and MoGe-based elevation mapping, integrating clearance-aware path planning with foothold-level plant avoidance.
 - Led greenhouse experiments and data analysis, achieving up to 0.94 segmentation IoU, 2.30 cm elevation MAE, and an average 56.8% reduction in vegetation contact across different vegetation-density scenarios.

Development of a Bionic Hexapod Robot with Adaptive Gait and Clearance for Agricultural Field Scouting
 - Designed and built a lightweight bionic hexapod robot platform for agricultural field scouting, featuring adjustable body clearance from 6–18 cm and an 8 kg payload capacity.
 - Developed a terrain-adaptive gait strategy that switches between low-clearance marching and high-clearance step-over modes for efficient obstacle crossing.
 - Led outdoor field experiments and Simulink simulations on slopes, grass, and obstacle-rich terrain, demonstrating stable locomotion on slopes up to 17° and a 16.2% reduction in obstacle-crossing energy cost compared with conventional obstacle avoidance.

EXTENSION AND MENTORING

- New Zealand SP Grower Visit:** Presented the robot design concept to the New Zealand Sweet Potato Association delegation and demonstrated the robot’s movement in the field and its precise pesticide spraying capabilities.
- Mentoring undergraduate students:** Mentored Cornell undergraduate student Kenneth Chenango Kniffin in designing the retractable nozzle mechanism for the 5-DOF leg, helping develop his interest in robotic mechanical design.

SKILLS

- Programming:** Python, L^AT_EX, C++, MATLAB, R
- Software:** SolidWorks, MATLAB, IsaacSim
- Hardware:** Laser Cutting Machine, 3D Printer (FDM and SLA)

HONORS AND AWARDS

- First Prize (Team Leader), 2024 ASABE Robotics Student Design Competition** July. 2024
- Best Prototype Award, NC State Sustainability Innovation Competition Make-A-Thon** Jan. 2024
- Outstanding Student Leader of Xi’an Jiaotong University** June. 2023
- Second-Class Scholarship, Xi’an Jiaotong University** Oct. 2022
- First Prize (Team Leader), 10th National College Students Mechanical Innovation Design Competition** June 2022